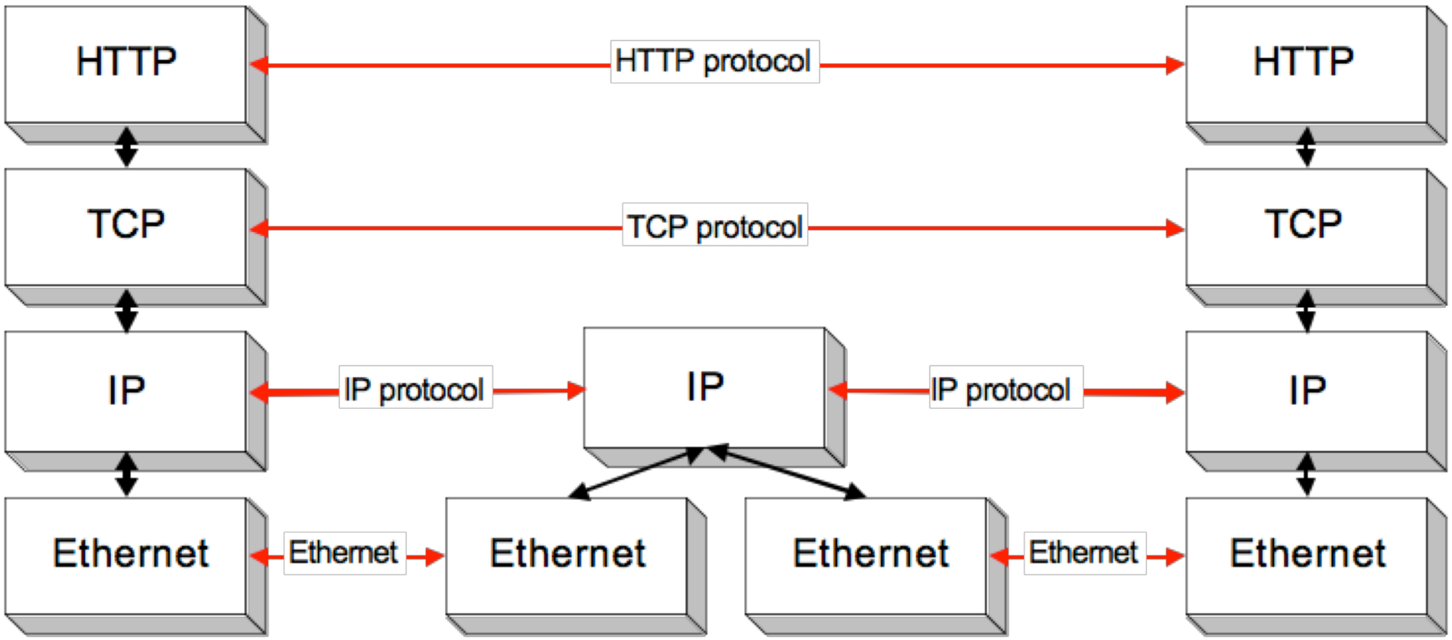
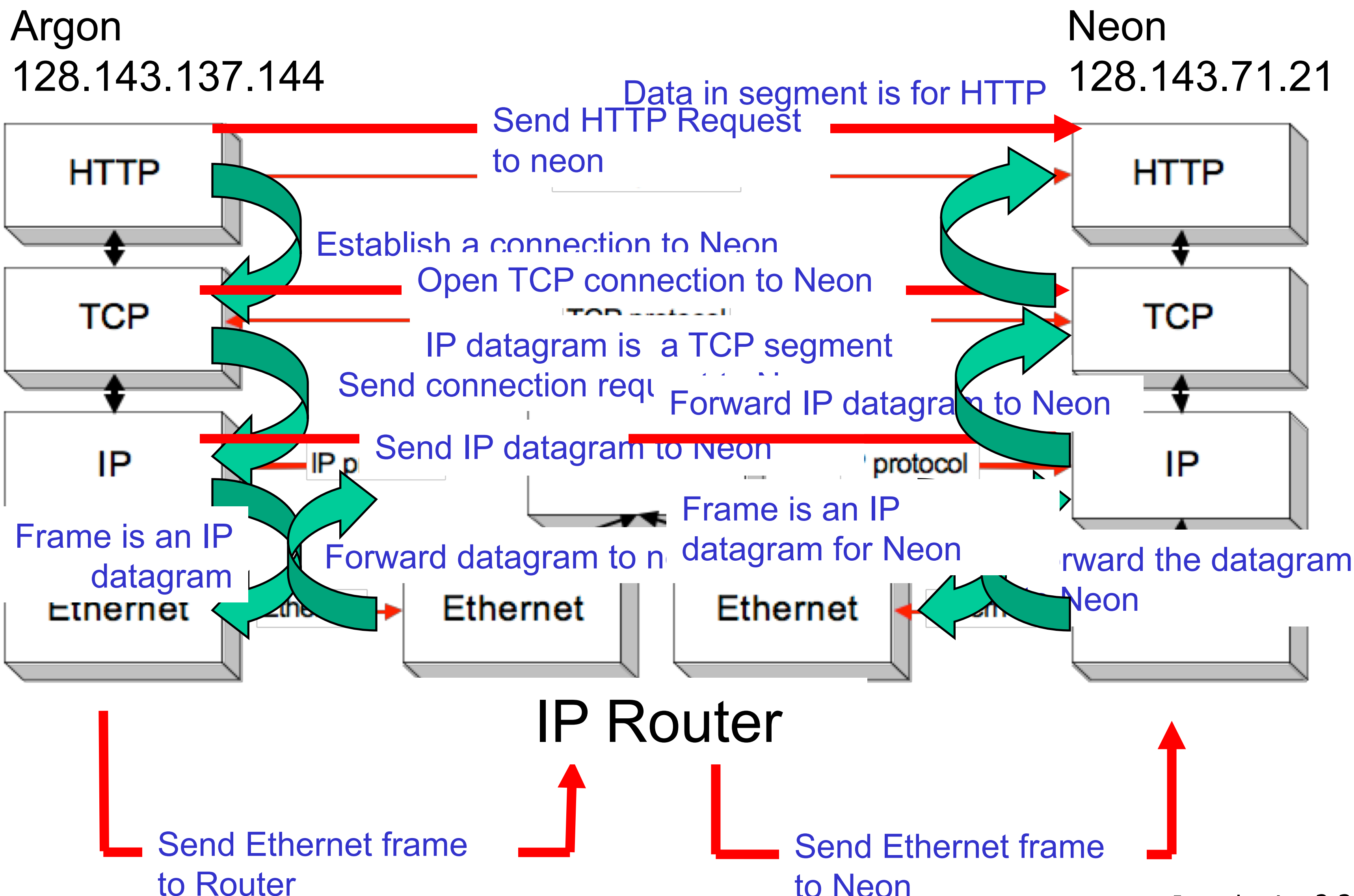
IEFT Protocol Stack Example



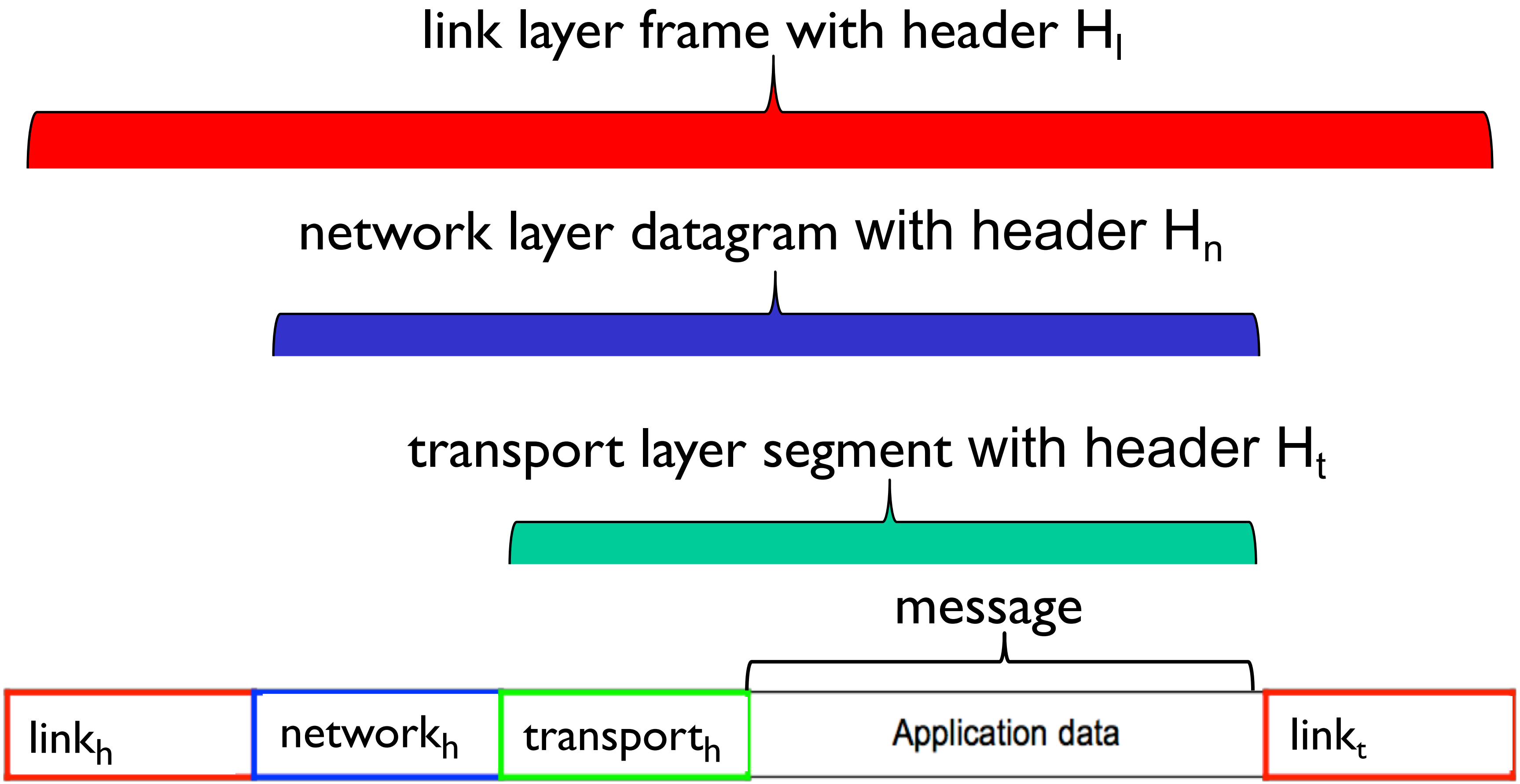
Protocols in Action



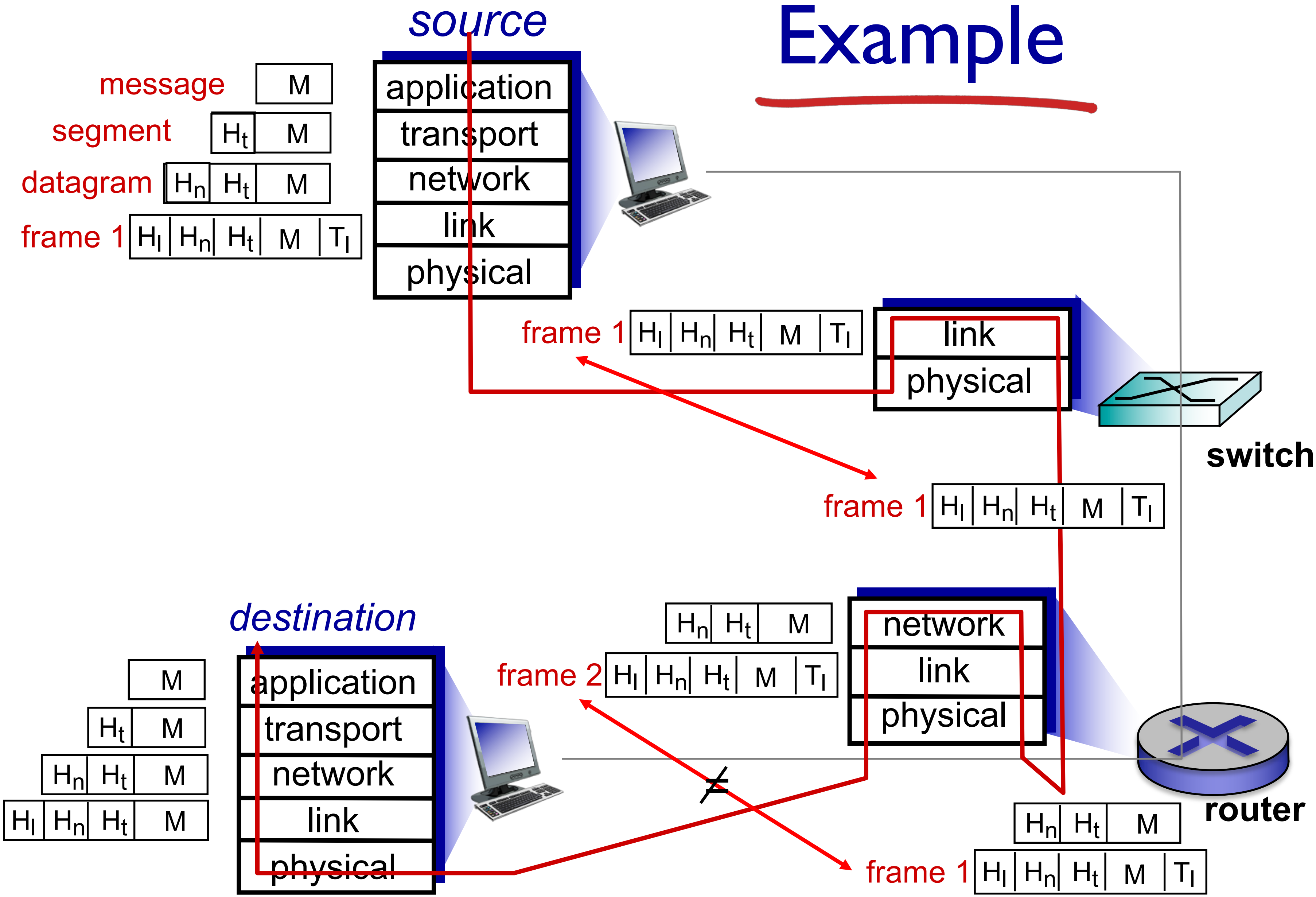
Encapsulation

- with layering comes encapsulation
- each layer adds control bits used to process the information to provide the desired type of service to the layer above it and ultimately to the application
- the packet grows in size as it drops from application to datalink at the source
- the packet is stripped of bits as it travels back up the stack to the application layer

IETF Encapsulation



Example



source
message M
segment H_t M
datagram H_n H_t M
frame 1 H_l H_n H_t M T_l
application
transport
network
link
physical
frame 1 H_l H_n H_t M T_l
link
physical
switch
frame 1 H_l H_n H_t M T_l
destination
M
H_t M
H_n H_t M
H_l H_n H_t M
application
transport
network
link
physical
H_n H_t M
frame 2 H_l H_n H_t M T_l
$\neq$
network
link
physical
router
H_n H_t M
frame 1 H_l H_n H_t M T_l

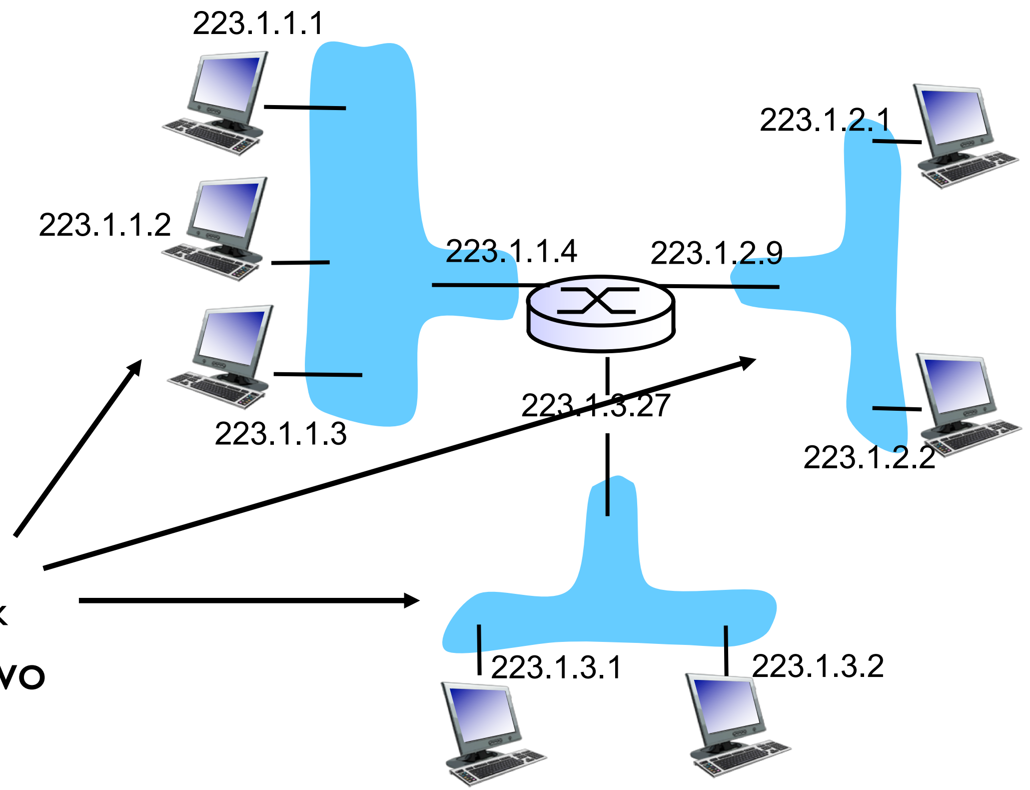
Chapter 4: Network Layer

4.3 IP: Internet Protocol

- IPv4 addressing

Internet Addressing: Introduction

- *Internet address:* 32-bit identifier for host, router *interface*
- *interface:* connection between host/router and physical link
  - router's typically have multiple interfaces
    - each interface ***creates*** a distinct/separate network
  - host typically has one or two interfaces (e.g., wired Ethernet, wireless 802.11)
    - interfaces may or may not be on a same network
- *Internet address associated with each interface*



$$223.1.1.1 = \underbrace{11011111}_{223}\ \underbrace{00000001}_{1}\ \underbrace{00000001}_{1}\ \underbrace{00000001}_{1}$$

What is an Internet address?

- Internet address is usually referred to as the IP (Internet Procotol) address.
- henceforth we will refer to it as IP address
- an IP address is a unique global address for a network interface
- an IP address:
  - is a 32 bit long identifier – 4bytes
  - encodes a network number referred to as network prefix and a device number referred to as host address
- human analogy:
  - network prefix → street name
  - host number → house number

IP Dotted Decimal Notation

- IP addresses are written in a dotted decimal format
- each byte is identified by a decimal number in the range [0….255] – 2^8
- example

10000000	10001111	10001001	10010000
1st Byte	2nd Byte	3rd Byte	4th Byte
= 128	= 143	= 137	= 144

128.143.137.144

Network prefix and host number

- the network prefix identifies a network and the device number identifies a specific device (actually, *interface* on the network as a device can have more than 1 ”network card”, and each card will have a unique IP address).
- real life analogy? - street name and house number

network prefix	**host number**

- **how do we know how long the network prefix is?**
  - the network prefix **used** to be implicitly defined (**class-based addressing, A,B,C,D…**)
  - the network prefix now is **flexible** and is indicated by a **prefix/netmask (classless).**

Example

Example: argon.cs.virginia.edu

- IP address is 128.143.137.144
  - is that enough info to route/forward a datagram??? -> No.
  - must indicate a network prefix for every IP device (host and router)
- using prefix notation an IP address is: 128.143.137.144/<u>x</u>
  - e.g., **x = 16** means network prefix is16 bits long
- the prefix is identified by a network mask: prefix =16 → mask consists of 16 'one's
- i.e., mask is represented as: **<u>255.255</u>.0.0** or hex format: **ffff0000**

--> **network prefix (or ID or address)** (IP address **AND** netmask) is: **128.143.0.0**

--> **host number** (IP address **AND** **inverse** of netmask(=**0000ffff**) is: **0.0.137.144**

128.143	137.144